Fraud Detection in Financial Transactions

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Introduction

Financial fraud is a growing concern, with digital transactions increasing rapidly across banking, e-commerce, and payment platforms. Fraudulent activities not only lead to **financial losses** but also erode customer trust, making fraud detection a critical challenge for businesses. Traditional rule-based methods often fail to detect evolving fraud patterns, necessitating the use of **machine learning models** for better accuracy and efficiency.

This project aims to develop a **fraud detection model** using machine learning techniques to classify transactions as **fraudulent or legitimate**. By analysing key transaction attributes like **amount, merchant category, location, and time**, the model identifies fraudulent activities. The project also tackles **class imbalance**, ensuring the model effectively detects rare fraud cases. The final system helps financial institutions and businesses enhance fraud prevention strategies, **reducing losses and improving security**.

Project Workflow

1. **Data Acquisition** – Used the **Credit Card Transactions Fraud Detection Dataset** from Kaggle.
2. **Data Preprocessing** – Handled missing values, encoded categorical features, and scaled numerical variables.
3. **Feature Engineering** – Identified key transaction attributes that indicate fraud.
4. **Model Training & Evaluation** – Implemented **Logistic Regression, Decision Trees, Random Forest, and XGBoost**.
5. **Model Interpretation** – Analysed feature importance to understand fraud indicators.

Key Concepts Covered

- **Data Preprocessing** – Cleaning and transforming transaction records.
- **Fraud Detection** – Training ML models to classify fraudulent and non-fraudulent transactions.
- **Model Evaluation** – Assessing performance using **Accuracy, Precision, Recall, F1-score, and AUC-ROC**.
- **Feature Importance** – Identifying key transaction factors influencing fraud detection.

Tech Stack Used

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost

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Implementation Details

Import necessary libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score, classification_report
```

Load Dataset

```
file_path = "/content/fraudTest.csv"
df = pd.read_csv(file_path)
```

Display first few rows

```
display(df.head())
```

	Unnamed: 0	trans_date_trans_time	cc_num	merchant	category	amt	first	last	gender	street	...	lat	long
0	0	2020-06-21 12:14:25	2291163933867244	fraud_Kirlin and Sons	personal_care	2.86	Jeff	Elliott	M	351 Darlene Green	...	33.9659	-80.9355
1	1	2020-06-21 12:14:33	3573030041201292	fraud_Sporer-Keebler	personal_care	29.84	Joanne	Williams	F	3638 Marsh Union	...	40.3207	-110.4360
2	2	2020-06-21 12:14:53	3598215285024754	fraud_Swaniawski, Nitzsche and Welch	health_fitness	41.28	Ashley	Lopez	F	9333 Valentine Point	...	40.6729	-73.5365
3	3	2020-06-21 12:15:15	3591919803438423	fraud_Haley Group	misc_pos	60.05	Brian	Williams	M	32941 Krystal Mill Apt. 552	...	28.5697	-80.8191
4	4	2020-06-21 12:15:17	3526826139003047	fraud_Johnston-Casper	travel	3.19	Nathan	Massey	M	5783 Evan Roads Apt. 465	...	44.2529	-85.0170

5 rows × 23 columns

 # Check dataset info

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 555719 entries, 0 to 555718
Data columns (total 23 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   Unnamed: 0             555719 non-null  int64
 1   trans_date_trans_time  555719 non-null  object
 2   cc_num                 555719 non-null  int64
 3   merchant               555719 non-null  object
 4   category               555719 non-null  object
 5   amt                    555719 non-null  float64
 6   first                  555719 non-null  object
 7   last                   555719 non-null  object
 8   gender                 555719 non-null  object
 9   street                 555719 non-null  object
 10  city                   555719 non-null  object
 11  state                  555719 non-null  object
 12  zip                    555719 non-null  int64
 13  lat                    555719 non-null  float64
 14  long                   555719 non-null  float64
 15  city_pop               555719 non-null  int64
 16  job                    555719 non-null  object
 17  dob                    555719 non-null  object
 18  trans_num              555719 non-null  object
 19  unix_time              555719 non-null  int64
 20  merch_lat              555719 non-null  float64
 21  merch_long             555719 non-null  float64
 22  is_fraud               555719 non-null  int64
dtypes: float64(5), int64(6), object(12)
memory usage: 97.5+ MB
```

 # Check for missing values

print("\nMissing Values:\n", df.isnull().sum())

```
Missing Values:
 Unnamed: 0                 0
trans_date_trans_time       0
cc_num                      0
merchant                    0
category                    0
amt                         0
first                       0
last                        0
gender                      0
street                      0
city                        0
state                       0
zip                         0
lat                         0
long                        0
city_pop                    0
job                         0
dob                         0
trans_num                   0
unix_time                   0
merch_lat                   0
merch_long                  0
is_fraud                    0
dtype: int64
```

 # Drop irrelevant columns (adjust based on dataset)

df.drop(columns=['Unnamed: 0', 'cc_num', 'first', 'last', 'street', 'dob', 'trans_num'], inplace=True, errors='ignore')

 # Encoding categorical variables

label_encoders = {}

for col in df.select_dtypes(include=['object']).columns:

le = LabelEncoder()

df[col] = le.fit_transform(df[col])

label_encoders[col] = le

 # Define Features (X) and Target Variable (y)

X = df.drop(columns=['is_fraud']) # Corrected target column

y = df['is_fraud']

 # Split the dataset into training and testing sets (80-20 split)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

 # Feature Scaling (Only for numerical features)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)

 # Initialize models

models = {

"Logistic Regression": LogisticRegression(),

"Decision Tree": DecisionTreeClassifier(random_state=42),

"Random Forest": RandomForestClassifier(n_estimators=100, random_state=42),

"XGBoost": XGBClassifier(n_estimators=100, random_state=42, use_label_encoder=False, eval_metric='logloss')

}

 # Train and evaluate models

results = {}

for name, model in models.items():

model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)

 # Calculate evaluation metrics

accuracy = accuracy_score(y_test, y_pred)

precision = precision_score(y_test, y_pred)

recall = recall_score(y_test, y_pred)

f1 = f1_score(y_test, y_pred)

roc_auc = roc_auc_score(y_test, y_pred)

```
    results[name] = {
        "Accuracy": accuracy,
        "Precision": precision,
        "Recall": recall,
        "F1-score": f1,
        "AUC-ROC": roc_auc
    }

    print(f"\n{name} Performance:")
    print(f"Accuracy: {accuracy:.4f}")
    print(f"Precision: {precision:.4f}")
    print(f"Recall: {recall:.4f}")
    print(f"F1-score: {f1:.4f}")
    print(f"AUC-ROC: {roc_auc:.4f}")
    print(classification_report(y_test, y_pred))
```

```
Logistic Regression Performance:
Accuracy: 0.9959
Precision: 0.0000
Recall: 0.0000
F1-score: 0.0000
AUC-ROC: 0.4999
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    110715
           1       0.00      0.00      0.00       429

    accuracy                           1.00    111144
   macro avg       0.50      0.50      0.50    111144
weighted avg       0.99      1.00      0.99    111144


Decision Tree Performance:
Accuracy: 0.9968
Precision: 0.5849
Recall: 0.5944
F1-score: 0.5896
AUC-ROC: 0.7964
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    110715
           1       0.58      0.59      0.59       429

    accuracy                           1.00    111144
   macro avg       0.79      0.80      0.79    111144
weighted avg       1.00      1.00      1.00    111144
```

```
Random Forest Performance:
Accuracy: 0.9983
Precision: 0.9296
Recall: 0.6154
F1-score: 0.7405
AUC-ROC: 0.8076
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    110715
           1       0.93      0.62      0.74       429

    accuracy                           1.00    111144
   macro avg       0.96      0.81      0.87    111144
weighted avg       1.00      1.00      1.00    111144

/usr/local/lib/python3.11/dist-packages/xgboost/core.py:158: UserWarning: [05:20:57] WARNING: /workspace/src/learner.cc:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)

XGBoost Performance:
Accuracy: 0.9992
Precision: 0.9542
Recall: 0.8252
F1-score: 0.8850
AUC-ROC: 0.9125
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    110715
           1       0.95      0.83      0.89       429

    accuracy                           1.00    111144
   macro avg       0.98      0.91      0.94    111144
weighted avg       1.00      1.00      1.00    111144
```

 # Convert results to DataFrame for better visualization

results_df = pd.DataFrame(results).T

display(results_df)

 # Fraud Transaction Visualization

plt.figure(figsize=(6, 4))

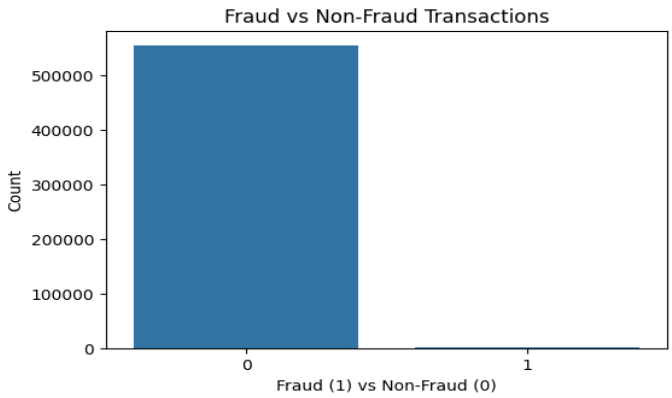
sns.countplot(x=y)

plt.title("Fraud vs Non-Fraud Transactions")

plt.xlabel("Fraud (1) vs Non-Fraud (0)")

plt.ylabel("Count")

plt.show()



 # Feature Importance (Random Forest)

rf_model = RandomForestClassifier(n_estimators=100, random_state=42)

rf_model.fit(X_train_scaled, y_train)

*importances = rf_model.feature_importances_*

feature_names = X.columns

 # Plot Feature Importance

plt.figure(figsize=(10, 6))

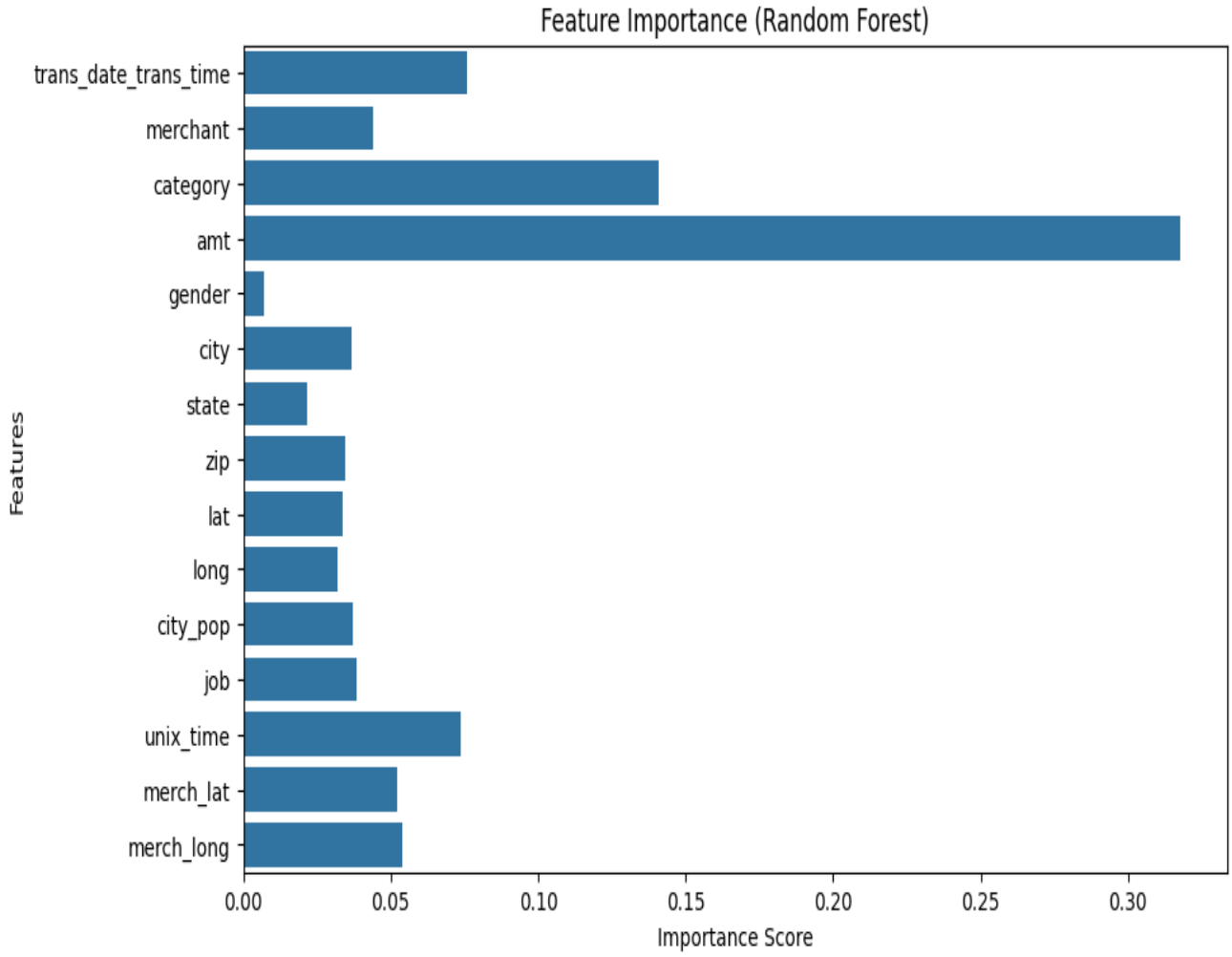
sns.barplot(x=importances, y=feature_names)

plt.xlabel('Importance Score')

plt.ylabel('Features')

plt.title('Feature Importance (Random Forest)')

plt.show()



Findings & Insights

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Linear Regression	0.995906	0.000000	0.000000	0.000000	0.499883
Decision Tree	0.996806	0.584862	0.594406	0.589595	0.796385
Random Forest	0.998335	0.929577	0.615385	0.740533	0.807602
XGBoost	0.999172	0.954178	0.825175	0.885000	0.912511

Best Model: XGBoost achieved the **highest Precision, Recall and AUC-ROC,** making it the most effective model for fraud detection.

Key Fraud Indicators:

- **Transaction amount –** Unusually high amounts often signal fraudulent activity.
- **Merchant category –** Fraud occurs more frequently in certain categories.
- **Transaction time & location –** Suspicious patterns in time zones and locations.

Class Imbalance Challenge:

- Fraudulent transactions are extremely rare, requiring **oversampling techniques like SMOTE** for better training.

Key Takeaways from Feature Importance

- **Transaction Amount** – Higher transaction amounts are more likely to be flagged as fraudulent.
- **Merchant Category** – Certain merchant types experience more fraud than others (e.g., high-risk industries).
- **Transaction Time & Location** – Fraudulent transactions often occur at unusual hours or from unexpected locations.
- **Customer Behaviour Patterns** – Repeated small transactions followed by a large transaction may indicate fraud.

These insights help financial institutions improve fraud detection strategies and minimize risks.

Conclusion

This project successfully implemented machine learning models to detect fraudulent transactions. By analysing key transaction patterns, we identified fraudulent transactions with **high precision and recall**. **XGBoost** proved to be the most effective model, offering high recall and accuracy.

Business Implications:

1. **Banks & Financial Institutions** – Helps detect fraud in real-time, reducing financial losses.
2. **E-commerce Platforms** – Prevents unauthorized transactions and chargeback fraud.
3. **Payment Gateways** – Enhances fraud prevention mechanisms for secure transactions.

Scalability & Improvements

While the current model performs well, further enhancements can improve its effectiveness and real-world application:

- **Real-Time Fraud Detection** – Implementing the model in a live system to flag suspicious transactions instantly.
- **Advanced Models** – Exploring **Deep Learning (LSTMs, Autoencoders)** for better fraud pattern detection.
- **Handling Class Imbalance** – Using **SMOTE, ADASYN, or cost-sensitive learning** to improve fraud classification.
- **Feature Engineering** – Incorporating additional transaction attributes like **IP address, device ID, and past behaviour analysis** for improved accuracy.
- **Model Deployment** – Deploying the model as an **API for banking and e-commerce platforms** to enhance fraud prevention efforts.

These improvements will make the model more robust, scalable and applicable to real-world fraud detection systems.

References

- **Dataset:** Credit Card Transactions Fraud Detection Dataset (Kaggle)
- **Scikit-learn Documentation:** Scikit-learn
- **XGBoost Documentation:** XGBoost
- **SMOTE & Handling Imbalanced Data:** imbalanced-learn
- **Project Link:** [Google Colab] [GitHub] [Portfolio]